1 Prompt Encoding & Overlapped Latent Initialization

P1 *A hot air balloon, early sunrise glow.*

P2 *A hot air balloon, bright daylight.*

PM *A hot air balloon, violet dusk light.*

Frozen Text Encoder(s)

C_1 C_2 C_M

$c_1, \ldots, c_M \Rightarrow \{c_s\}_{s=1}^{S}$ *with transition conditions*

Initial Gaussian Noise $\epsilon_s \sim N(0, I)$

Segment 1 Segment 2 Segment S

Noise Reuse
(tail overlap of s → head overlap of s+1)

Latent block ($T \times C \times H' \times W'$) | Overlap region($O$) | Gaussion noise

2 Iterative Denoising with FreqSync (repeat for $n = N, N-1, \ldots, 1$)

Input at step n
noisy latents z_n + text conditions $\{c_s\}_{s=1}^{S}$

A FS-SCA (inside selected attention layers $\ell \in \mathcal{L}_{FS-SCA}$)

Target Self-Conditioned Path
Q_t → K_t, V_t → Y_{tar}

Source-Conditioned Path (cross-attention)
Q_t → Aligned K_s, V_s → Y_{src}

Frequency Routing (temporal FFT)
Low (structure) / Mid (appearance) / High (details)
$\lambda_{low} > \lambda_{mid} > \lambda_{high}$

Boundary Gate $G(\rho, n, \ell)$

Fuse (blend)

Residual Clipped Fused Output Y_{out}

Residual Clip

$$Y_{out} = Y_{tar} + clip_r(Y_{sync} - Y_{tar}; Y_{tar})$$

Predicted Denoised Latents $\{\hat{z}_0^{(s)}(n)\}_{s=1}^{S}$

B RCSS (after prediction, before sampler update)

$$\hat{Z}^{guard} = Z^{guard} + clip_{r_g}(\tilde{Z}^{guard} - Z^{guard}; Z^{guard})$$

Prev-tail / Cur-head overlap predictions

Temporal FFT (along time)

Spectral Stitching
High-Freq (preserve details) | Low-Freq (align & smooth)

Residual Clipping

Stitched Overlap

Sampler Update
$z_{n-1}^{(s)} = Update(z_n^{(s)}, \hat{z}_0^{(s)}(n), n)$

3 Overlap Removal/ Concatenation & VAE Decoding

Clean latents $\{z_0^{(s)}\}_{s=1}^{S}$

Remove overlaps & concatenate

VAE Decoder (frozen)

Final Long Video

flow of data | main data flow | gating / modulation | residual clipping | depends on boundary Position ρ, step n, layer ℓ

ρ: normalized position inside the boundary window
n: denoising step / noise level
ℓ: attention layer index